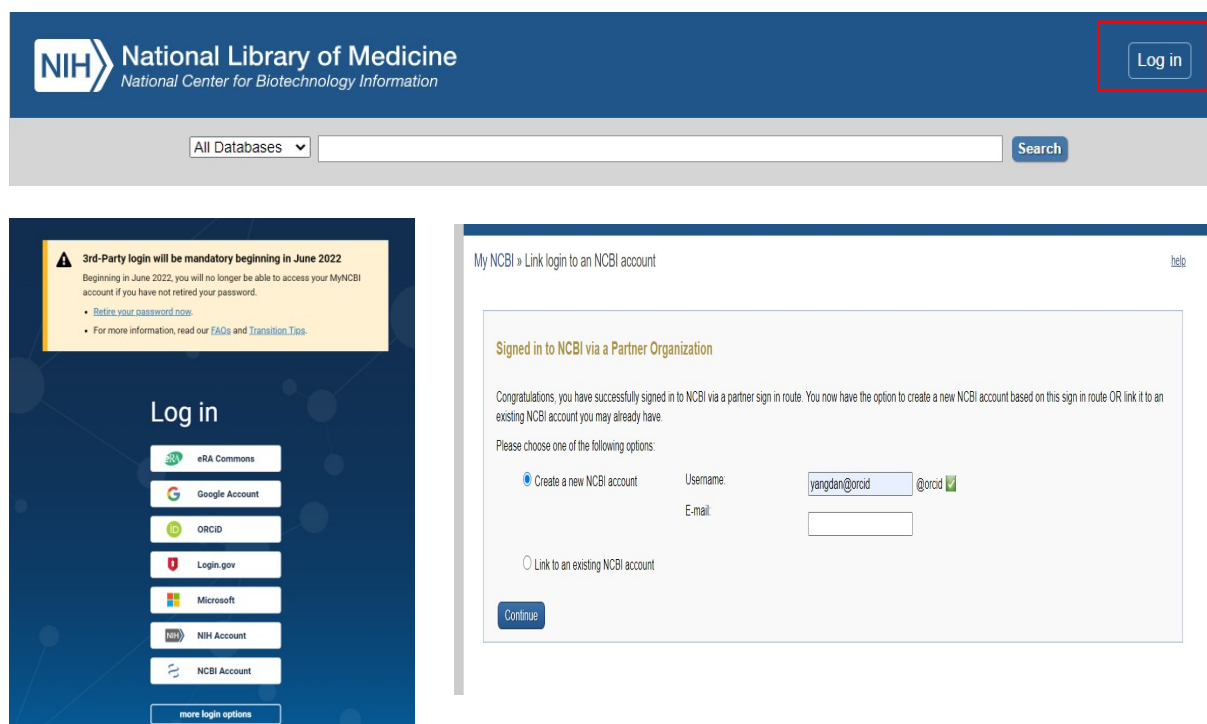


GEO 数据上传操作指南

一、GEO 数据上传操作方法

1、NCBI 账号注册

数据上传前需要注册一个 NCBI (<https://www.ncbi.nlm.nih.gov/>) 账号，创建需要邮件激活。如果已经有账号，可以直接点击登录。



2、GEO 账号注册

Getting Started	GEO Tools	More Resources
GEO Documentation	Submit to GEO	GEO Home
GEO FAQ	Advanced Search	GEO Profiles
About GEO DataSets	DataSet Browser	SRA
Construct a Query	Programmatic Access	
Download Options	GEO2R	
Example Searches		




[GEO Publications](#) | [FAQ](#) | [MIAME](#) | [Email GEO](#)

[NCBI](#) » [GEO](#) » [Info](#) » [Submitting data](#)
[Login](#)

Submitting data

GEO accepts many categories of high-throughput functional genomic data, including all array-based applications and some high-throughput sequencing data.

[NCBI](#) » [GEO](#) » [Submitter](#)

My GEO Profile

WARNING: Submitters with a qq.com, 163.com, or foxmail.com email address do not always receive mail from NCBI. Please provide another email address from a different domain to ensure that we are able to communicate with you.

Investigator

Use this section to provide details about the primary investigator. This information will be displayed on GEO records.

First name*

Middle name

Last name*

You may choose not to display your email or phone on GEO records by unchecking the corresponding checkbox.

E-mail(s)*
1034149180@qq.com

Phone

Show e-mail ☒

Show phone ☒

Organization name*

Department

Lab

Street address*

City*

State/province

ZIP/Postal code*

Country*

Submitter (Account manager)

If the person responsible for submitting the data to GEO is different from the Investigator, use this section to provide alternative contact information. This situation typically arises when the submitter is, e.g., microarray facility personnel, but the contact details to display with the data are that of the principal investigator.

Both the Investigator and the Submitter will receive e-mail correspondence from GEO.

If the Submitter contact details are the same as the Investigator information, leave the "Same as Investigator" checkbox checked.

Same as Investigator ☒

Name

E-mail(s)

Phone

Preview how contact information will be displayed on GEO records. Edits to contact information will be applied immediately to all existing GEO records submitted under that account.

[Preview](#)
[Save](#)
[New submission](#)

HHS Vulnerability Disclosure

NCBI 账号关联 GEO 数据库提交账号

填写姓名、电话、E-mail 等基本信息（带*号的为必填信息）

注：qq.com、163.com 或 foxmail.com 电子邮件地址可能不能收到来自 NCBI 的邮件。请提供其他邮箱以确保 NCBI 能够与您沟通。

注册成功后会收到 GEO 注册成功邮件。

注：GEO 帐户三个月内不上传数据将自动删除。

3、数据上传导航




[GEO Publications](#) | [FAQ](#) | [MIAME](#) | [Email GEO](#)

[NCBI](#) » [GEO](#) » [Info](#) » [Submitting data](#)
User: 1034149180@qq.com | [My submissions](#) | [Logout](#)

Submitting data

GEO accepts many categories of high-throughput functional genomic data, including all array-based applications and some high-throughput sequencing data.

We aim to make data deposit procedures as straightforward as possible and will provide as much assistance as you require to get your data submitted to GEO. If you have problems or questions about submission, e-mail us with a brief description of the type of data you are trying to submit, and one of our curators will quickly get back to you.

Data types

- Submit microarray
- Submit high-throughput sequencing **RNA-seq 选择高通量测序**
- Submit other (includes NanoString, RT-PCR, traditional SAGE)

WARNING: If you are submitting human data, it is your responsibility to comply with Human Subject Guidelines.

All Databases

[NCBI Home](#)
[Resource List \(A-Z\)](#)
[All Resources](#)
[Chemicals & Bioassays](#)
[Data & Software](#)
[DNA & RNA](#)
[Domains & Structures](#)
[Genes & Expression](#)
[Genetics & Medicine](#)
[Genomes & Maps](#)
[Homology](#)
[Literature](#)
[Proteins](#)

[All Databases](#)
[Assembly](#)
[Biocollections](#)
[BioProject](#)
[BioSample](#)
[Books](#)
[ClinVar](#)
[Conserved Domains](#)
[dbGaP](#)
[dbVar](#)
[Gene](#)
[Genome](#)
[GEO DataSets](#)
[GEO Profiles](#)
[GTR](#)
[HomoloGene](#)
[Identical Protein Groups](#)
[MedGen](#)
[MeSH](#)
[NCBI Web Site](#)


Submit
 Upload your manuscripts
 bases


Download
 Transfer NCBI data to your
 computer


Learn
 Find help documents, attend a
 class or watch a tutorial

NCBI
 Center for Biotechnology Information advances science and health by providing access to
 genomic information.
[Mission](#) | [Organization](#) | [NCBI News & Blog](#)

Popular Resources
[PubMed](#)
[Bookshelf](#)
[PubMed Central](#)
[BLAST](#)
[Nucleotide](#)
[Genome](#)
[SNP](#)
[Gene](#)
[Protein](#)
[PubChem](#)

NCBI News & Blog

1) STUDY 主要包含文章的标题、概述、实验整体设计、共同作者等

# This is a metadata template for submission of high-throughput sequencing samples to the GEO database. # Fields with a red triangle in the upper-right corner have pop-up instructions that can be displayed if you hover over the field. # HINT: For bulk RNA-Seq processed data, please provide a tab-delimited matrix or individual text files reporting complete abundance measurements (eg, raw and/or normalized data used as input for differential expression analysis).	
STUDY	GSE131369
# This section describes the overall study.	
# Information provided in this section will be displayed in a GEO Series (GSE record) on public web pages.	
title	Effect of depletion of KDM4D, NFIB, and MLL1 on gene expression during adipogenic differentiation of C3H10T1/2 mesenchymal stem cells
summary (abstract)	To investigate the cooperative function KDM4D/NFIB/MLL1 complex in the regulation of adipogenic differentiation, we established 10T1/2 cell lines in which each target gene was depleted.
summary (abstract)	We then performed gene expression profiling analysis using data obtained from RNA-seq of 4 different cells at two time points.
experimental design	Comparative gene expression profiling analysis of RNA-seq data for C3H10T1/2 cells and its KD derivatives (shKDM4D, shNFIB, and shMLL1).
contributor	Hansol Lee
contributor	Janghyun Choi
supplementary file optional	

2) SAMPLES: 包括样品名称、样品的详细信息、物种、细胞系、细胞类型、试验处理情况、单双端测序情况、测序机器、原始文件名等（目前公司均是使用 Illumina NovaSeq 6000 测序仪器，采用 paired-end 测序）

library name	title	organism	cell line	cell type	genotype	treatment	time	molecule	single or paired-end
Sample 1	C3H10T1/2 cells, shControl, Day0	Mus musculus	C3H 10T1/2	mesenchymal stem cells	WT	adipogenic differentiation	Day 0	total RNA	paired-end
Sample 2	C3H10T1/2 cells, shControl, Day5	Mus musculus	C3H 10T1/2	mesenchymal stem cells	WT	adipogenic differentiation	Day 5	total RNA	paired-end
Sample 3	C3H10T1/2 cells, shKDM4D, Day0	Mus musculus	C3H 10T1/2	mesenchymal stem cells	KDM4D knockdown	adipogenic differentiation	Day 0	total RNA	paired-end
Sample 4	C3H10T1/2 cells, shKDM4D, Day5	Mus musculus	C3H 10T1/2	mesenchymal stem cells	KDM4D knockdown	adipogenic differentiation	Day 5	total RNA	paired-end
Sample 5	C3H10T1/2 cells, shNFIB, Day0	Mus musculus	C3H 10T1/2	mesenchymal stem cells	NFIB knockdown	adipogenic differentiation	Day 0	total RNA	paired-end
Sample 6	C3H10T1/2 cells, shNFIB, Day5	Mus musculus	C3H 10T1/2	mesenchymal stem cells	NFIB knockdown	adipogenic differentiation	Day 5	total RNA	paired-end
Sample 7	C3H10T1/2 cells, shMLL1, Day0	Mus musculus	C3H 10T1/2	mesenchymal stem cells	MLL1 knockdown	adipogenic differentiation	Day 0	total RNA	paired-end
Sample 8	C3H10T1/2 cells, shMLL1, Day5	Mus musculus	C3H 10T1/2	mesenchymal stem cells	MLL1 knockdown	adipogenic differentiation	Day 5	total RNA	paired-end

instrument model	description	processed data file	processed data file	raw file	raw file	raw file
Illumina NovaSeq 6000	control	gene_FPKM.txt		shControl_D0_1.fq.gz (释放数据中raw data read1文件名)	shControl_D0_2.fq.gz (释放数据中raw data read2文件名)	
Illumina NovaSeq 6000	control	gene_FPKM.txt		shControl_D5_1.fq.gz	shControl_D5_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shKDM4D_D0_1.fq.gz	shKDM4D_D0_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shKDM4D_D5_1.fq.gz	shKDM4D_D5_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shNF1b_D0_1.fq.gz	shNF1b_D0_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shNF1b_D5_1.fq.gz	shNF1b_D5_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shMLL1_D0_1.fq.gz	shMLL1_D0_2.fq.gz	
Illumina NovaSeq 6000	treatment (具体的处理方式)	gene_FPKM.txt		shMLL1_D5_1.fq.gz	shMLL1_D5_2.fq.gz	

Processed data files 主要是基因表达的数据文件（readcount 或 FPKM），需要换成文本 txt 格式。

raw file 对应原始数据 raw data 中的压缩包名称，以 fq.gz 结尾，由于是双端测序，因此一个样本对应-1、-2 两端数据。

3) PROTOCOLS: 样本的处理方案，建库方案，类型，处理步骤，处理过程中文件信息等

PROTOCOLS						
# Information provided in this section will appear in each GEO Sample (GSM record). These fields can be moved to the above SAMPLES section in order to provide different information for different samples.						
# Please ensure that the annotations are relevant to the SAMPLES above (not to the samples from a previous submission).						
growth protocol optional	C3H 10T1/2 mesenchymal cells were maintained in DMEM supplemented with 10% fetal bovine serum (FBS) and antibiotics in humidified atmosphere with 5% CO2 at 37C					
treatment protocol optional	Adipogenic differentiation was induced by treatment of 2-day post confluent cells (designated day 0, D0) with 10ug/ml insulin (Sigma).					
treatment protocol optional	1uM dexamethasone (Sigma), and 0.5mM isobutylmethylxanthine (Sigma) in the presence of 10% FBS until day 2. Cells were then fed with DMEM supplemented with 10% FBS and 10ug/ml insulin for 2 days, after wh					
extract protocol	RNA was harvested using RNeasy mini plus kit (Qiagen). 1.3 ug of total RNA was used for the construction of sequencing libraries.					
library construction protocol	RNA libraries for RNA-seq were prepared using SMARTER mRNA-Seq Library Prep Kit following manufacturer's protocols.					
library strategy	RNA-seq					
data processing step	CLC Genomics Workbench v11.0.1					
data processing step	Sequence reads were trimmed for adaptor sequence/low-quality sequence using CLC genomic benchmark (parameter- Quality limit: 0.05)					
data processing step	Trimmed sequence reads were mapped to GRCm38/mm10 using CLC genomic benchmark (parameters- mismatch cost: 2 insertion cost: 3 deletion cost: 3 length fraction: 0.8 similarity fraction: 0.8)					
data processing step	Read count extraction and normalization were performed using CLC genomic benchmark					
genome build/assembly	mm10					
processed data files format and content	tab-delimited text files include RPKM values for each Sample					

4) PAIRED-END EXPERIMENTS: 双端测序的样本 fq 文件名

PAIRED-END EXPERIMENTS						
# If "paired-end" experiments are included, list the files for each paired-end run in a row. Each row will become one sequencing run on a GEO Sample (GSM record).						
# Single-cell data: If applicable, list index files (i1, i2, etc.) in "file name 3", "file name 4" columns.						
# Please make sure all raw files listed here are also listed in the "raw file" columns in the above SAMPLES section.						
file name 1			file name 2			
shControl_D0_1.fq.gz (释放数据中raw data read1文件名)			shControl_D0_2.fq.gz (释放数据中raw data read2文件名)			
shControl_D5_1.fq.gz			shControl_D5_2.fq.gz			
shKDM4D_D0_1.fq.gz			shKDM4D_D0_2.fq.gz			
shKDM4D_D5_1.fq.gz			shKDM4D_D5_2.fq.gz			
shNF1b_D0_1.fq.gz			shNF1b_D0_2.fq.gz			
shNF1b_D5_1.fq.gz			shNF1b_D5_2.fq.gz			
shMLL1_D0_1.fq.gz			shMLL1_D0_2.fq.gz			
shMLL1_D5_1.fq.gz			shMLL1_D5_2.fq.gz			

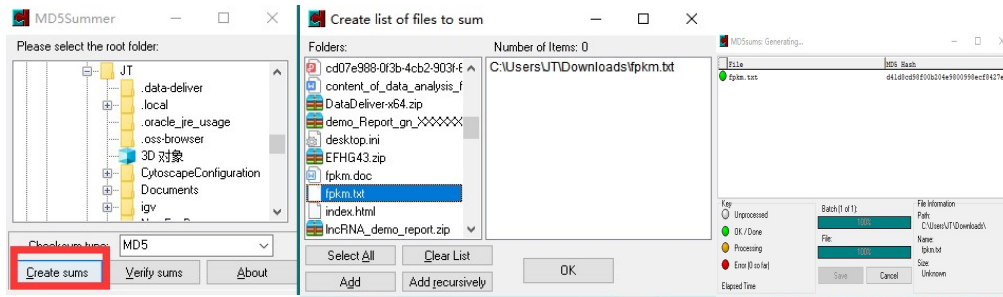
5) MD5 Checksums

MD5 即 Message-Digest Algorithm 5，是当前计算机领域广泛使用的哈希算法之一，用于确保信息传输完整一致。

# We strongly recommend that submitters provide MD5 checksums for their raw data files. The checksums are used to verify file integrity. Checksums can be calculated using the following methods:						
# Unix: md5sum <file>						
# OS X: md5 <file>						
# Windows: Application required. Many are available for free download.						
RAW FILES			PROCESSED DATA FILES			
file name	file checksum		file name	file checksum		
Capan_1_1.fq.gz	7200e9db4ce7541f225b4634c5e1d6d7		pkm.txt	d41d8cd98f00b204e9800998ecf8427e		
Capan_1_2.fq.gz	43c76bf31edb03ecbf7dd338d760558b					

RAW FILES 即 raw data，对应的 file checksum 可直接在释放的全部数据中的 md5.txt 直接查看。

PROCESSED DATA FILES 即结果文件中的表达量 fpkm 文件，需将文件转换成 txt 格式后上传，windows 电脑的 file checksum 可利用 md5.exe（释放的全部数据中有）生成，具体操作如下。

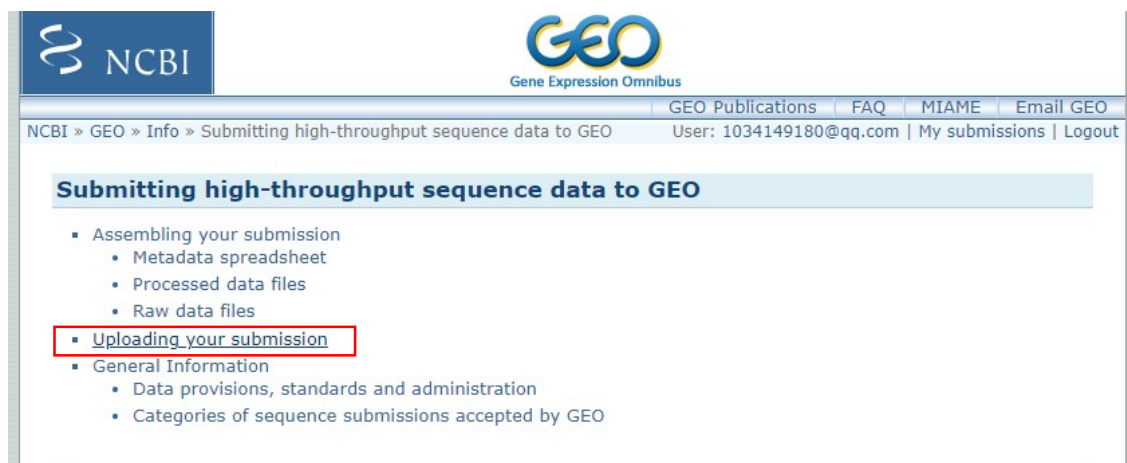


5、数据上传

将以上涉及的三部分（表格与数据）打包到一个文件夹内

注：可接受的压缩格式是 gzip 和 bzip2（即以 .gz 或 .bz2 扩展名结尾的文件）。切勿压缩二进制文件（例如，BAM、bigWig、bigBed），也不要上传 ZIP 档案（扩展名为 .zip 的文件）。

1) 进入数据上传界面



Uploading your submission

There are two steps for submission:

1. Transfer all your files to the GEO FTP server.

If your files are >1 terabyte, please contact us and do not transfer your files until you hear back from us.

Transfer Files

2. After the FTP transfer is complete, notify GEO using the Submit to GEO web form

Notify GEO

GEO File Transfer Protocol (FTP)

Step 1. Your personalized upload space is: **uploads/yangdan202206@126.com_2fKRMCAg**

上传路径

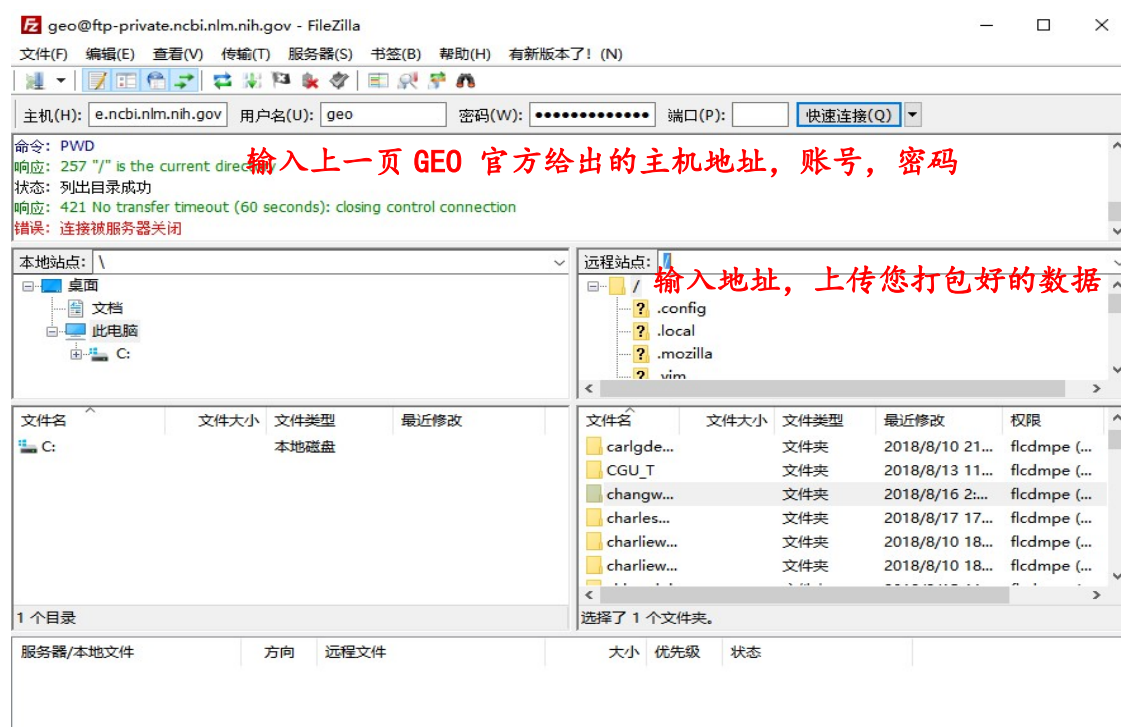
需要下载该软件进行数据传输

- geo_submission_maintenance_seq.
- Confirm the size of your folder and **contact us if your submission exceeds 1 terabyte in size**. Please do not proceed with an upload larger than 1 terabyte until you hear back from GEO.
 - For PC/Mac OS users we recommend transferring files with the free third-party software, [FileZilla Client](#). Please see below for detailed examples and other options.
 - For LINUX/UNIX users, we recommend transferring files with 'ncftp' or 'lftp', but you can also use 'ftp', 'sftp', or 'ncftput'. Please see below for detailed examples.
 - Our FTP server credentials are:

host address	ftp-private.ncbi.nlm.nih.gov
username	geoftp
password	rebUzyi1

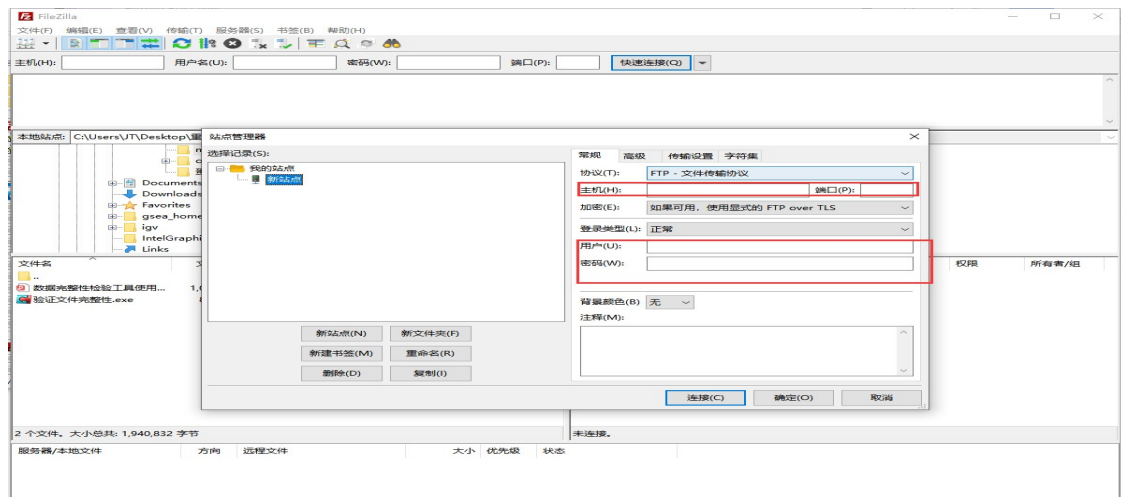
上传的主机，账号及密码

“快速连接”工具栏中输入主机(ftp-private.ncbi.nlm.nih.gov)，用户名(geoftp)和密码(rebUzyi1)来快速连接，会看到“快速连接”错误，可以忽略此错误。在“远程站点”地址栏中输入个性化工作区的路径（如我的路径为：/ uploads / zhaoyuhuan_8zOL842G）。然后通过从“本地站点”窗口拖动包含所有提交文件的文件夹并将其拖放到右侧的上传空间（“远程站点”窗口）中来传输文件。

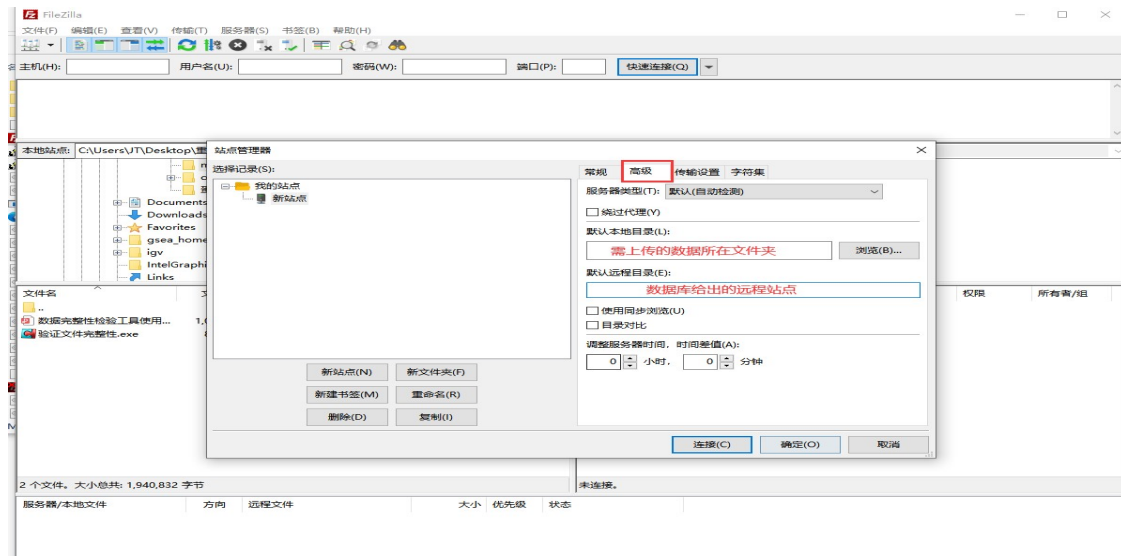


也可以通过设置站点管理器避免目标列表错误，步骤如下：

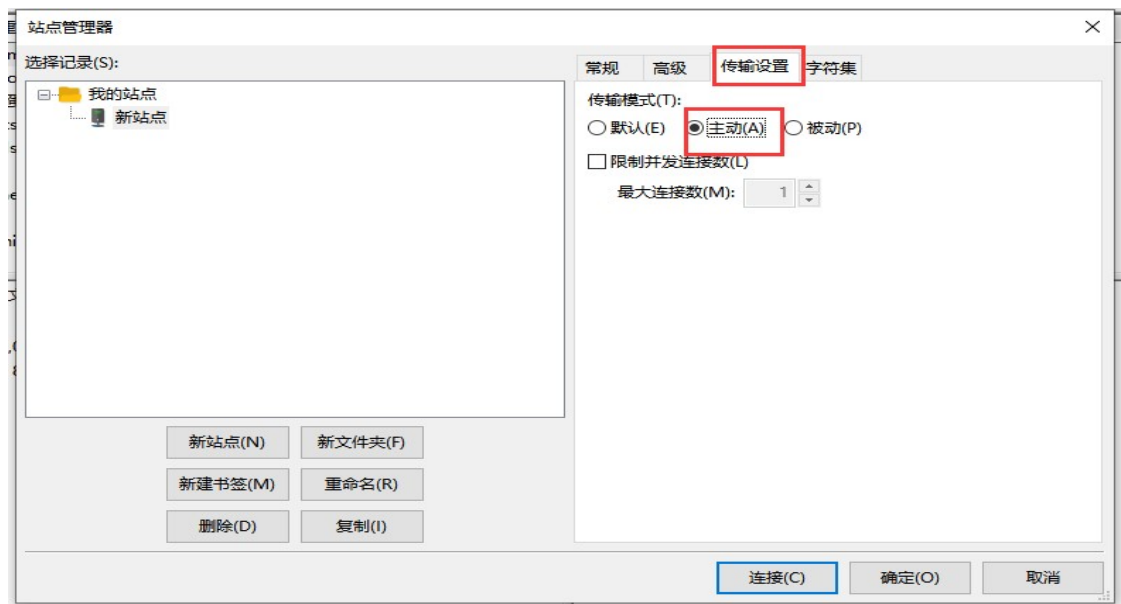
- 1) 选择文件-站点管理器-常规-输入主机（ftp-private.ncbi.nlm.nih.gov），用户名（geoftp）和密码（rebUzyi1）



2) 选择文件-站点管理器-高级-选择需要上传数据的文件夹-数据库给出的远程站点



3) 选择文件-站点管理器-传输设置-选择主动-点击连接即可



6、邮件确认

待传输完成后，点击 Notify GEO 通知 GEO 审核数据，GEO 官方大概在 5 个工作日左右回复，在收到管理员邮件确认之前不要引用 GEO 号。

Step 3. After the FTP transfer is complete, notify GEO using the Submit to GEO web form

Notify GEO

You must notify us after uploading your files. If you fail to notify us your files will be automatically deleted from the server after two weeks. Once notified, we move files to a safe location for review.

NCBI GEO Home Documentation Query & Browse Email GEO wangbaojian My NCBI Sign Out My GEO Submissions

Submit to GEO

Use this form to either:

- ☒ Notify GEO about your FTP file transfer (suitable for high-throughput sequencing or large microarray submissions and updates)

Is your FTP file transfer to GEO complete?

☒ Yes, all my data have finished transferring

Do not proceed with this form until all components of your submission are fully transferred and ready for us to process.

Name(s) of the directory or files deposited

Submission kind

☒ new

☐ update or revision

When this submission should be released to the public

☐ Release immediately following curation

☐ Release on specified date (up to 3 years from today)

Comment to GEO staff (optional)

Submit

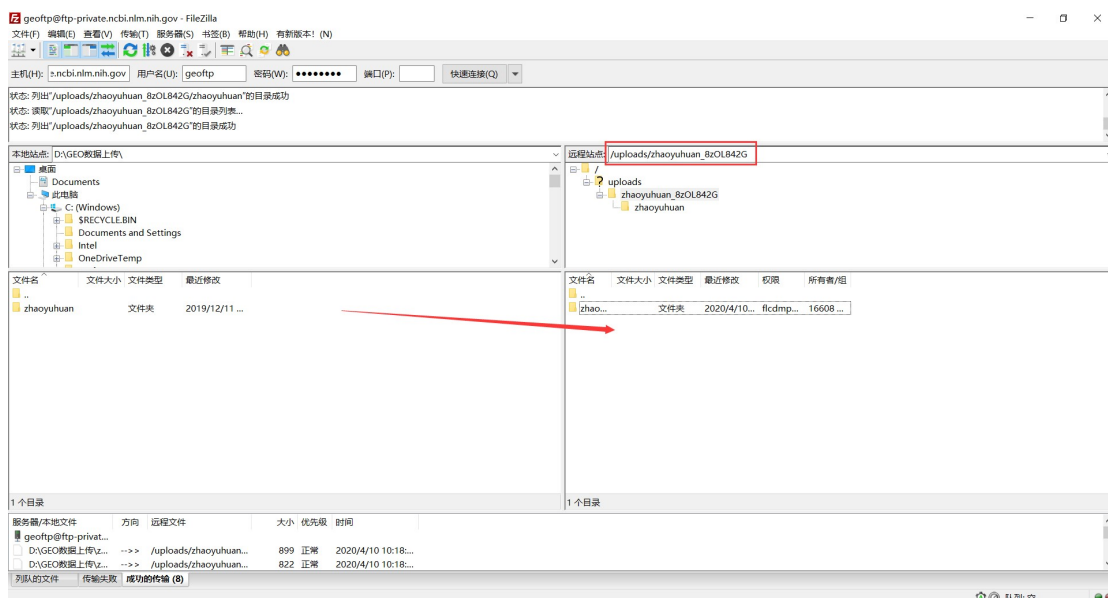
☐ No, I need help

注：若老师选择 3 年后公开，后续想更改时间，可以进行提交界面，通过设置修改时间，或者直接反馈 GEO 官网进行修改。

二、GEO 数据上传常见问题

1. FileZilla 连接失败问题

忽略错误，在远程站点处输入 GEO 给出的上传路径，回车后将要上传的文件夹拖曳到右侧即可开始上传。



2. FTP 上传数据报 550 错误（请求操作未被执行，文件不可用），可以尝试以下的方法来解决：

1) 准备要传输的文件，不要打开使用，很多人经常会忘记关闭了个别的文件，同时又将文件发给别人，这种正在使用着的文件是传输不过去的；

2) 文件的体积比较大，传输过程中耗费时间比较长，网络不稳定，容易造成传输文件中断，可以尝试将体积比较大的文件分开传输，或者尝试将文件压缩一些体积，打包传输；

3) 网络条件比较差，网络不给力，传输文件比较困难，可以在网络较好的情况下再传输；

4) 电脑上某些杀毒软件拦截文件，造成文件传输的错误，可以尝试暂时关闭特殊的杀毒软件。

3. Ftp 服务器连接失败，分为以下四种情况

1) 连接被拒，错误信息如下：

正在连接到 www.yourdomain.com -> DNS=www.yourdomain.com

IP=218.13.164.102 PORT=21

连接失败（连接被拒）

原因:这是因为客户在作 Ftp 上传时 填写错了 Ftp 服务器造成这个问题。

解决:在 Ftp 上传时服务器填写

3) .FTP 用户登入失败 错误信息如下：

正在连接到 www.72dns.com -> DNS=www.72dns.com IP=211.155.224.184

PORT=21 已连接到 www.72dns.com (Ftp 服务器连接成功)

220 Serv-U FTP Server v6.2 for WinSock ready...

USER test

331 User name okay, need password.

PASS (隐藏)

530 Not logged in.

原因:这是用户填写错误的 FTP 服务器、Ftp 用户名/或 Ftp 密码

解决:请您核对您的 Ftp 信息（开通空间时，系统会把 FTP 信息发到你邮箱里，你可以查看一下邮件）。如果您忘记 Ftp 密码。可以在“用户中心——虚拟主机管理——控制面板”里重设 Ftp 密码。

3). 用户本地上网问题，错误信息如下：

无法解析主机：s501.72dns.com

C:\>ping s501.72dns.com

Ping request could not find host s501.72dns.com.please check the name and try again.

原因:如这两种情况同时都出现，那是用户本地上网的 DNS 有问题。

解决:请用其它的 DNS 试下。

4). Ftp 服务器解析正常, Ftp 服务器连接超时，错误信息如下：

正在连接到 s501.72dns.com -> DNS=s501.72dns.com IP=203.171.239.16

PORT=21 连接失败（连接超时）

原因:网络方面有问题。

解决:更换网络再次尝试。

总结:目前最常见的 FTP 错误检查主要是检查 FTP 服务器、Ftp 用户名/或 Ftp 密码是否正确；切换站点管理器中传输设置的主动被动模式；站点管理器常

规中的加密选项选择只使用普通FTP（不安全）。